Forces I Practice

1. Force = mass x gravitational field strength

a) $F = 1500\cancel{kg}(9.8\frac{N}{\cancel{kg}}) = \boxed{15000\,N}$

b) $F = 4.5kg(9.8\frac{N}{kg}) = \boxed{44N}$

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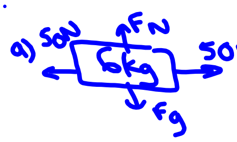
2. $F = mg$

a) $167N = m(9.8\frac{N}{kg}) \Rightarrow \boxed{m = 17.0kg}$

b) $408N = 250kg(g) \Rightarrow \boxed{g = 1.6\,\frac{N}{kg}}$

3.

a) 

• "constant speed" means $F_{net} = 0$

$0 = 50 - F_F \Rightarrow F_F = 50N = \mu F_N$

$F_g = mg$

$50 = \mu(6)(9.8)$

$\boxed{\mu = 0.85}$

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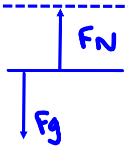
b) $F_F = \mu F_N = \mu mg = 0.35(5.0)(9.8) = \boxed{17N}$



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c) constant speed $\Rightarrow F_{net} = 0 \Rightarrow F_F = 880N = \mu mg$

$880 = \mu(100)(9.8) \Rightarrow \boxed{\mu = 0.9}$

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4. $F_e = kx$

a) $F_e = 40\frac{N}{m}(0.1m) = \boxed{4N}$

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b) $150N = 600\frac{N}{m}(x) \Rightarrow \boxed{x = 0.3m}$

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c) $2500N = k(0.5 - 0.4m) \Rightarrow \boxed{k = 25000\,\frac{N}{m}}$

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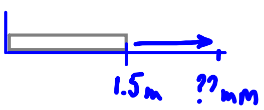
d) $6.0\times10^{3}\,N = 1.4\times10^{6}\frac{N}{m}(d_f - \underbrace{0.0015}_{d_i})$



$\frac{6.0\times10^3}{1.4\times10^6} + 0.0015 = \boxed{d_f = 5.8\,mm}$

5. $F_{net} = ma = \text{winner} - \text{loser}$

a) $F = (5.0\,kg)(2.0\,m/s^2) = 1.0 \times 10^1\,N$

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b) $12.0\,N = 0.030\,kg(a) \Rightarrow \boxed{a = 4.0 \times 10^2\,m/s^2}$

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c) $1000\,N = m(4.0 \times 10^3\,m/s^2) \Rightarrow \boxed{m = 0.3\,kg}$

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d) $20\,N = 5.0\,kg(a) \Rightarrow \boxed{a = 4\,m/s^2}$

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e) $150\,N = m(10\,m/s^2) \Rightarrow \boxed{m = 15\,kg}$

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6. a) $F_{net} = ma = \text{thrust} - \text{friction}$

$1200\,kg(a) = 700\,N - 500\,N \Rightarrow \boxed{a = 0.2\,m/s^2}$

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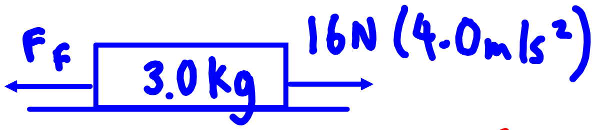
b)



$F_F = \mu F_N$

$(3.0\,kg)(4.0\,m/s^2) = 16\,N - \mu(3.0\,kg)(9.8\,\frac{N}{kg}) \Rightarrow \boxed{\mu = 0.14}$

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c) $m(2.8\,m/s^2) = 650\,N - 0.52\,m(9.8\,\frac{N}{kg})$

$m(2.8\,\frac{m}{s^2} + 0.52(9.8\,\frac{N}{kg})) = 650\,N \Rightarrow \boxed{m = 82\,kg}$

7. a) i) $F_g = 2.92\times10^6\,\cancel{kg}\left(9.8\frac{N}{\cancel{kg}}\right) = \boxed{2.9\times10^7\ N}$

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ii) $F_{net} = \text{thrust} - \text{gravity} = 2.4\times10^8 N - 2.9\times10^7 N = \boxed{2.1\times10^8 N}$

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iii) $F = ma \Rightarrow 2.1\times10^8 N = 2.92\times10^6 kg(a) \Rightarrow \boxed{a = 72\ m/s^2}$

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iv) fuel burns $\Rightarrow \underset{\text{constant}}{F} = m\downarrow a\uparrow$

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b) $F_{net} = ma = \text{thrust} - \text{gravity} \rightarrow F_g = mg$

i) $13140kg(a) = 2.63\times10^5 N - (13140kg)\left(9.8\frac{N}{kg}\right)$

$\boxed{a = 10.2\ m/s^2\ [up]}$

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ii) $4170kg(a) = 2.63\times10^5 N - 4170kg\left(9.8\frac{N}{kg}\right)$

$\boxed{a = 53.3\ m/s^2}$

8. a) $v_f = v_i + at$

$50 = 0 + a(0.05) \Rightarrow a = 1000\,m/s^2$

$F_{net} = ma = (5.0kg)(1000\,m/s^2)$

$\boxed{F_{net} = 5000\,N}$

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b) $F_{net} = ma \Rightarrow 48N = (50 + 10kg)a \Rightarrow a = 0.8\,m/s^2$

$v_f = v_i + at$

$4.0\,m/s = 0\,m/s + (0.8\,m/s^2)t \Rightarrow \boxed{t = 5\,s}$

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c) $F_{net} = F_g - F_{bouyancy}$

constant speed = balanced forces

$F_g = F_{bouyancy}$

$5.0kg(9.8\frac{N}{kg}) = \boxed{49N}$

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d) $v_f = v_i + at$

$48\frac{\cancel{km}}{\cancel{h}} \cdot \frac{\cancel{1h}}{3600s} \cdot \frac{1000m}{\cancel{1km}} = 0\,m/s + a(5.0\,s)$

$a = 2.7\,m/s^2$

$F_{net} = ma$

$2500N = m(2.7\,m/s^2)$

$\boxed{m = 940\,kg}$

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e) $6.6N = 9.0\,kg(a) \Rightarrow a = 0.73\,m/s^2$

i) $(3.0\,m/s)^2 = (0)^2 + 2(0.73\,m/s^2)d \Rightarrow \boxed{d = 6.1\,m}$

ii) $3.0\,m/s^2 = 0 + 0.73\,m/s^2(t) \Rightarrow \boxed{t = 4.1\,s}$